

Principles and advances in biostatistics

MASTER OF PUBLIC HEALTH PROGRAM

STAT 7311 - BIOSTATISTICS

LECTURE 1

Muhammad Adil Zainal Abidin

“Statistics is the grammar of science” *Karl Pearson*
(1857 – 1936, influential English mathematician) ”

STAT7311 Biostatistics

STAT7242 Advanced biostatistics

STAT8311 Causal inference in epidemiology

STAT 8212 Managing data in research

HAPM 8233 Health informatics

STAT7311

iTaleem

Attendance

Lecture notes material [github/adilzainal](https://github.com/adilzainal)

Assignment

Final exam

Practical - hybrid / online



R Practical

1. Download R <https://cran.r-project.org/>
2. Download R Studio <https://posit.co/downloads>
3. Download the practical dataset healthstatus.sav from Github
<https://github.com/adilzainal/>

Assignment and final exam

Attendance

Assignment - written statistical report (60%)

Final exam - Open book (40%) 3 hours

CLO

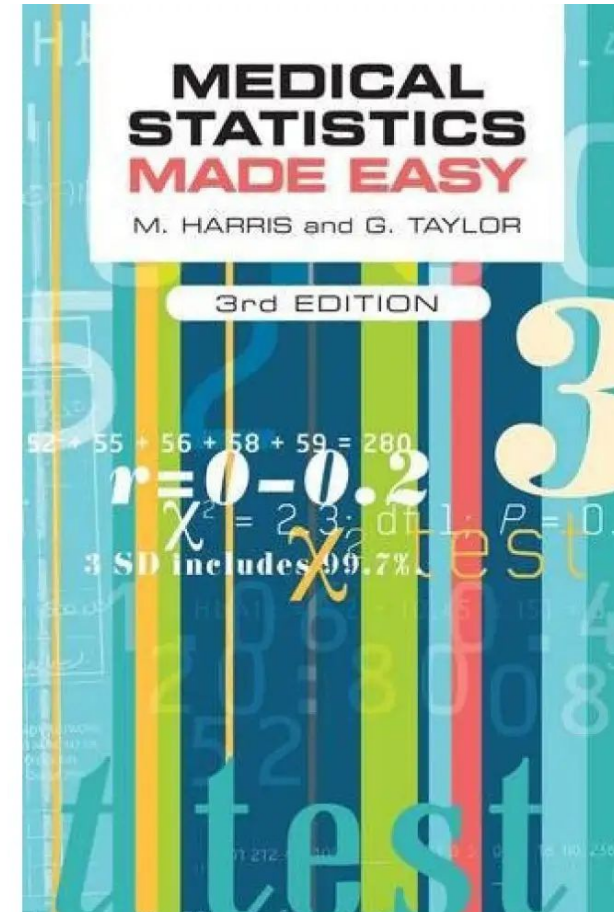
1. Describe the principles of biostatistics correctly based on the textbooks. (PLO 1)
2. Analyse statistical data using correct univariate or multivariate tests. (PLO 2)
3. Correctly interpret and report statistical analysis results. (PLO 3)
4. Construct a statistical analysis plan and execute it, individually or in a group. (PLO 4, PLO 5)

Topic Learning Outcome

1. Define biostatistics and explain its role in public health research.
2. Distinguish types of variables, data scales and common study summaries.
3. Recognise recent developments, ethical responsibilities and limitations in statistical practice.

JARGON

- Error type I and II
- Power
- Level of significance
- One sided vs two-sided test
- Effect size
- Sample size
- Clinical significance vs statistical significance
- Parameter vs statistics
- Variables



Biostatistics

“Application of statistical principles to questions and problems in medicine, public health or biology”



STATISTICS

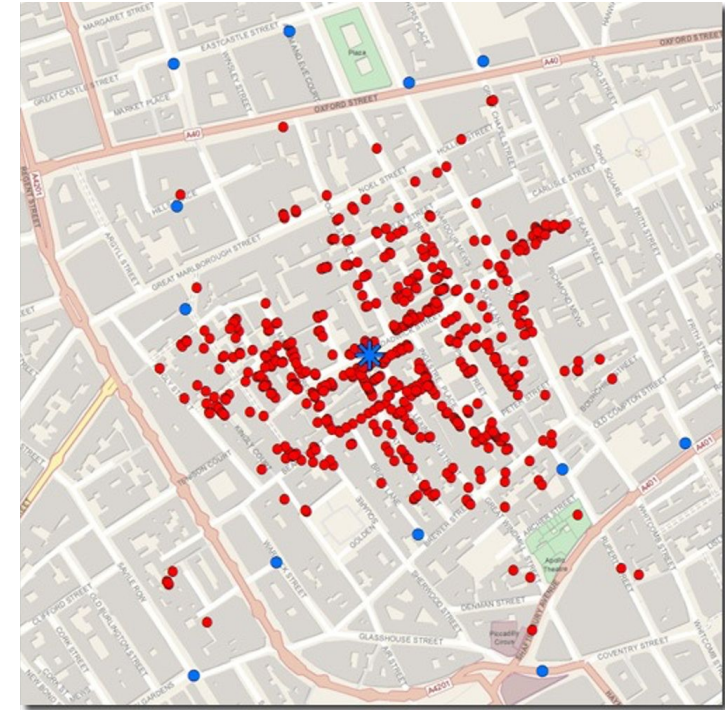
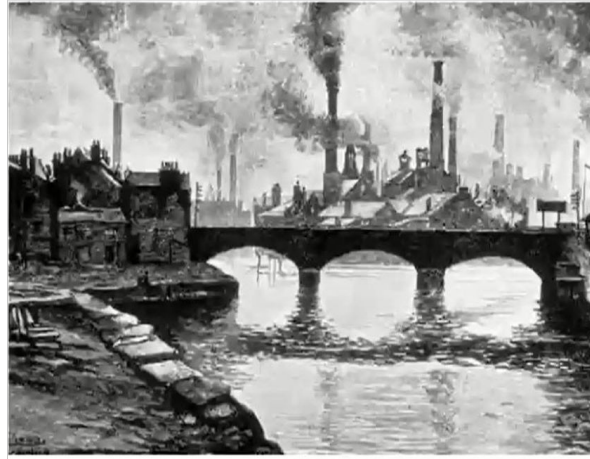
Statistics is a field of science concerned with :

1. Collection, organization, summarisation and analysis of data.
2. Drawing of inferences about a body of data when only a part of the data is observed. (Daniel W.)

Biostatistics / Medical statistics- Application of statistics in biological sciences and medicine.

Econometrics, Machine learning, Actuarial Science, Psychometrics

CHOLERA EPIDEMIC IN LONDON 1854



Ordinan Kuarantin 1868 di Negeri-Negeri Selat.

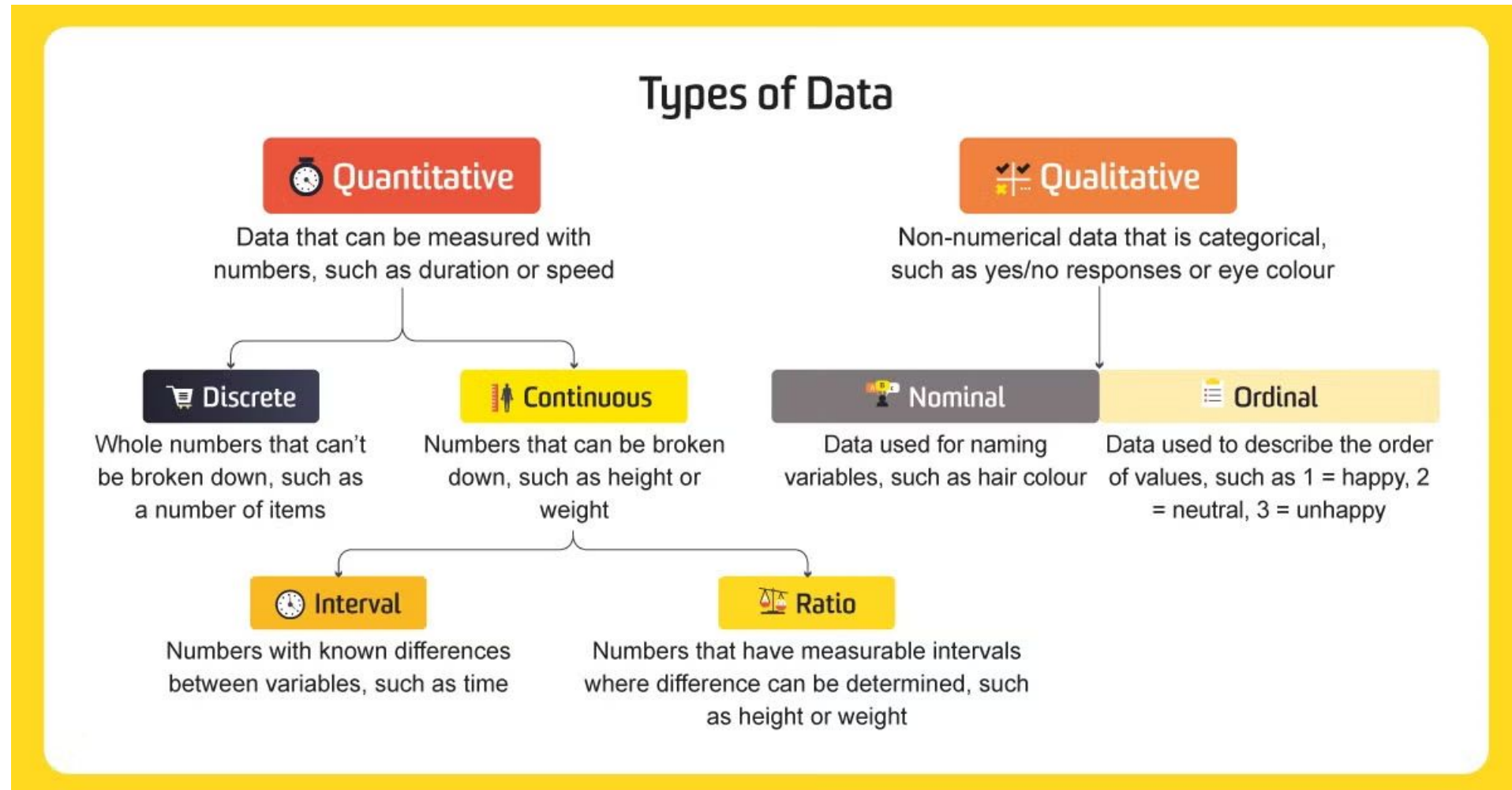
Water Company	# Houses	# Cholera Deaths	Death/10,000 Houses
Southwark & Vauxhall	40,046	1,263	315
Lambeth	26,107	98	37
Rest of London	256,423	1,422	59

Types of variables

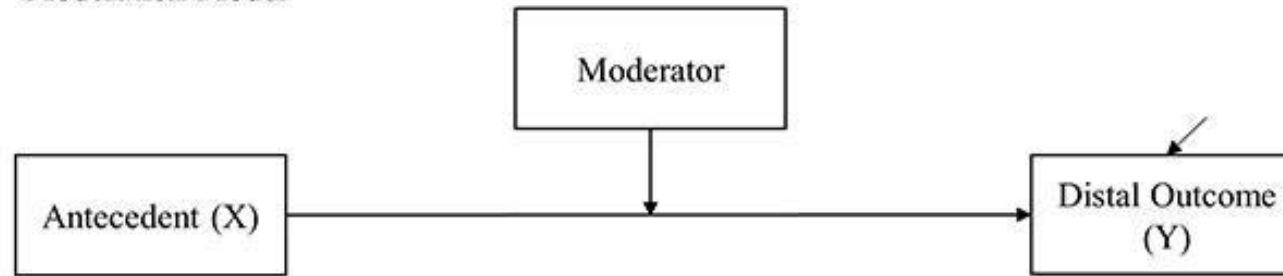
TYPES OF VARIABLES IN RESEARCH



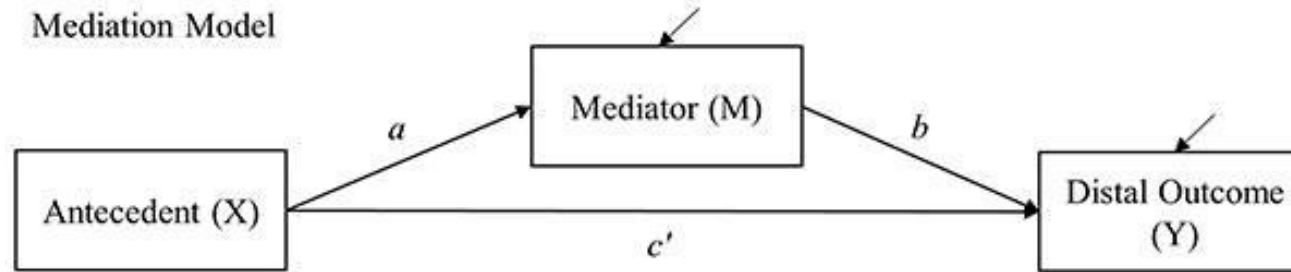
Levels of measurement



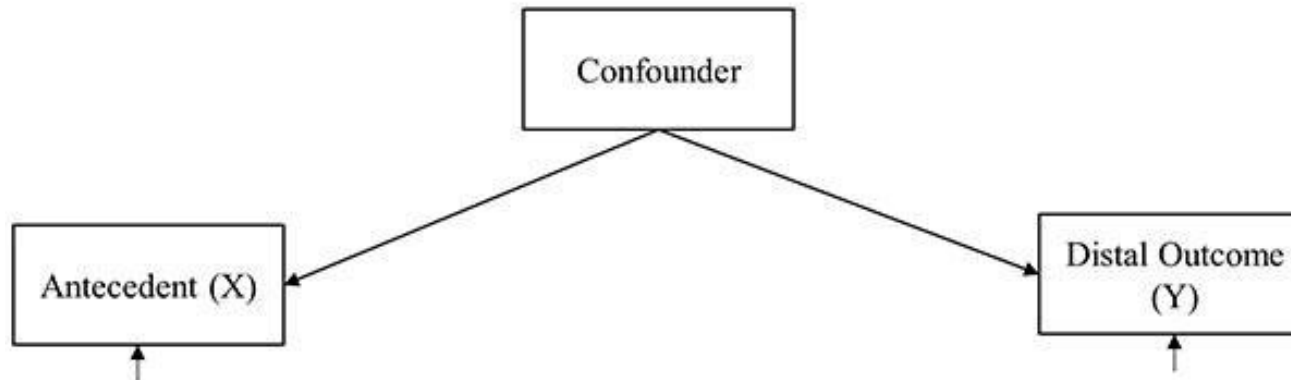
Moderation Model



Mediation Model

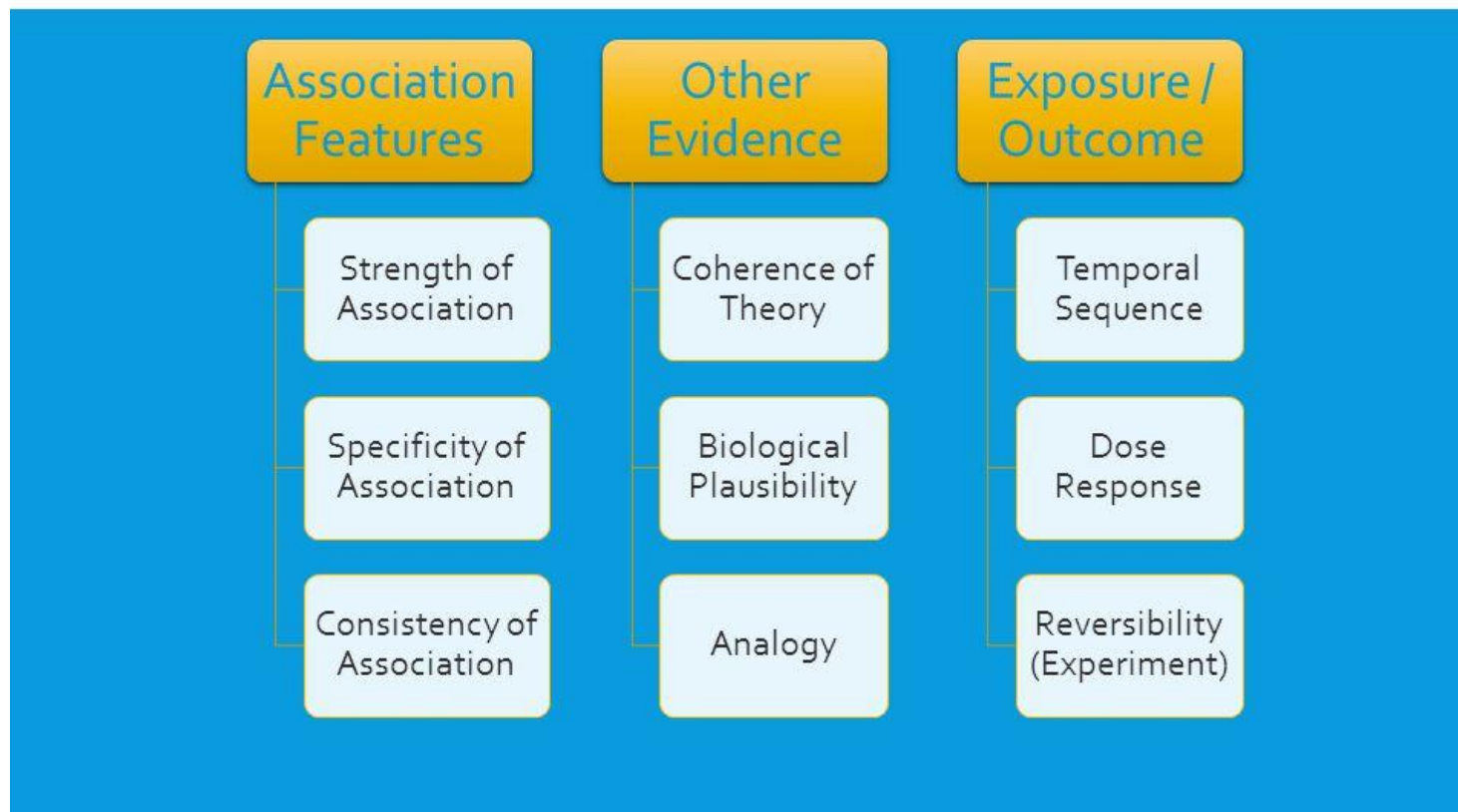


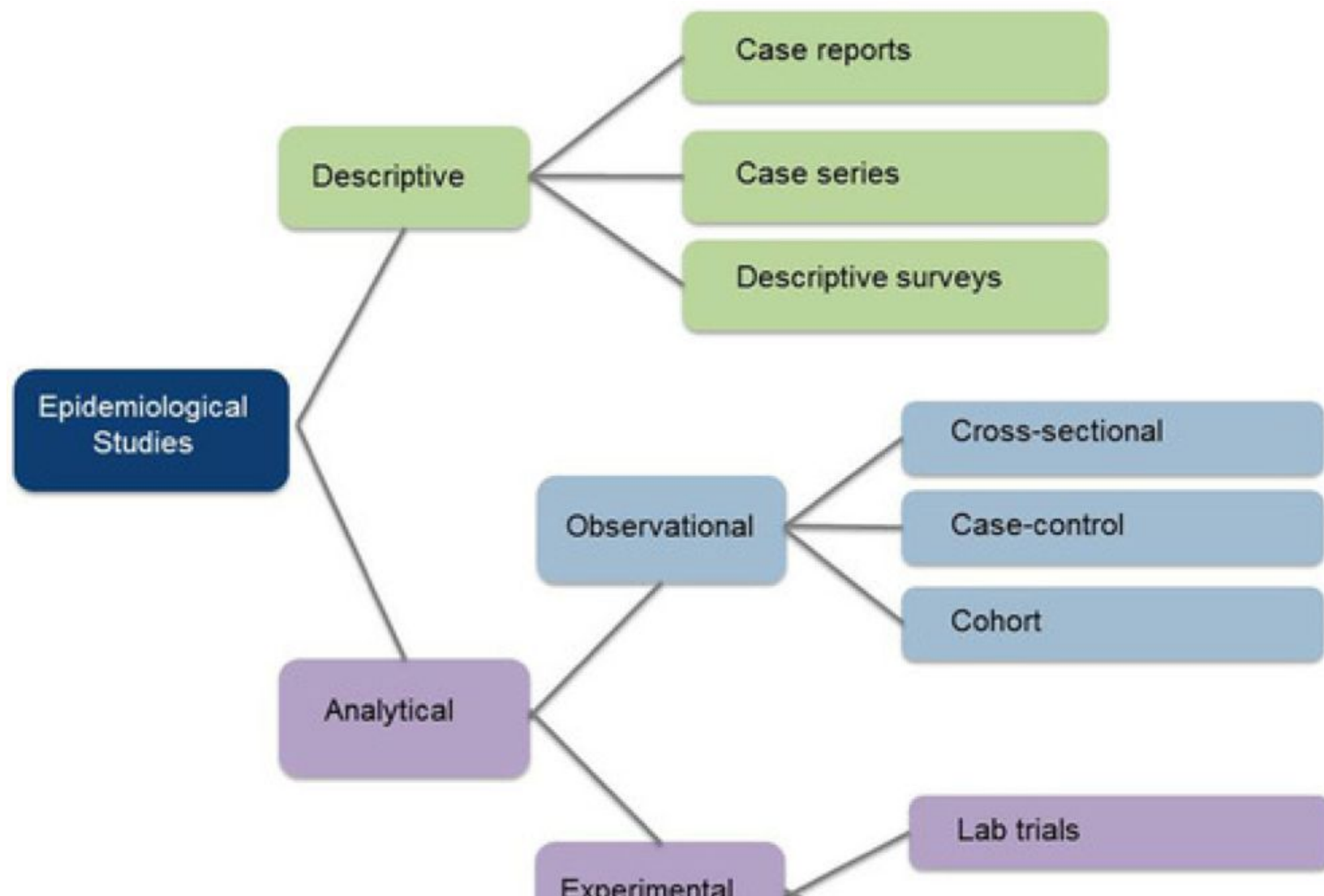
Confounding Model



Bradford Hill Criteria

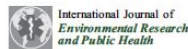
9 CRITERIA





CROSS SECTIONAL STUDY

- Measure Prevalence
 - No of new and old cases



Article

Prevalence and Determinants of Gestational Diabetes Mellitus: A Cross-Sectional Study in China

Xianglong Xu ^{1,2,3} , Ying Liu ⁴, Dengyuan Liu ^{1,2,3}, Xiaoming Li ^{1,2,3}, Yunshuang Rao ^{1,2,3}, Manoj Sharma ⁵ and Yong Zhao ^{1,2,3,*}

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Abstract: **Objectives:** This study aimed to identify the prevalence of gestational diabetes mellitus (GDM) and to examine its associations with social and behavioral factors, maternal body mass index (BMI), anemia, and hypertension. **Methods:** A cross-sectional analysis was performed on data collected from 2345 pregnant women from 16 hospitals in five selected provinces in mainland China. **Results:** Prevalence of GDM was as follows: overall: 3.7%; pregnant women in the first pregnancy: 3.4%; pregnant women in the second pregnancy: 4.6%. Compared with early pregnancy women, late-stage pregnant women were more likely to have GDM (OR = 4.32, 95% CI (1.82, 10.27)). Compared with 18–25 years old pregnant women, women aged 36–45 years were more likely to have GDM (OR = 3.98, 95% CI (1.41, 11.28)). Compared with non-hypertensive patients, hypertensive patients were more likely to have GDM (OR = 6.93, 95% CI (1.28, 37.64)). However, second pregnancy, high maternal BMI, prolonged screen time (TV-viewing time, computer-using time, and mobile-phone using time), insufficient and excessive sleep duration, poor sleep quality, smoking, and secondhand smoke exposure were not significantly associated with an increased risk of GDM. **Conclusions:** Women in the second pregnancy do not appear to predict an increased risk for developing GDM than women in the first pregnancy. High-risk groups of GDM included women in their late pregnancy, aged 36–45 years old, and with hypertension. The findings will contribute to an improved understanding of social and behavioral determinants of GDM in Chinese population and contribute to the development of health-prevention promotion interventions to address GDM.

Keywords: gestational diabetes mellitus; prevalence; social and behavioral determinants; China

$$\text{prev}A\text{Lence} = \frac{\text{ALL cases}}{\text{Population @ risk}}$$

$$iN\text{cidence} = \frac{\text{New cases}}{\text{Population @ risk}}$$

CASE CONTROL STUDY

- Identify risk factors – temporality, causality
- Measure odds ratio
- Odd of the risk factor
- Eg – Among first epidemiological study linking HPV and cervical cancer.

Int. J. Cancer: 52, 743-749 (1992)
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THE CAUSAL LINK BETWEEN HUMAN PAPILLOMAVIRUS AND INVASIVE CERVICAL CANCER: A POPULATION-BASED CASE-CONTROL STUDY IN COLOMBIA AND SPAIN

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¹Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ²Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ³Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁴Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁵Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁶Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁷Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁸Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ⁹Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ¹⁰Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France; ¹¹Unit of Field and Intervention Studies, International Agency for Research on Cancer, 150, Cours Albert Thomas, F-69372 Lyon, France.

METHODS

To evaluate the association between human papillomavirus (HPV) and cervical cancer, we performed a population-based case-control study in Colombia and Spain, the former country having an incidence rate of cervical cancer about 6 times higher than the latter. It included 438 cases of histologically confirmed invasive cervical cancer and 387 randomly selected population controls. Information on demographic variables, sexual behavior and other risk factors was obtained by interview. HPV-DNA was measured in cervical swabs specimens with 3 hybridization assays: Virapap[®], Southern hybridization (SH) and polymerase chain reaction (PCR). The presence of HPV-DNA was measured in each country regardless of the detection of HPV. In both countries combined, the adjusted odds ratio (OR) for HPV was 25.9 (95% confidence interval (CI) = 10.0-66.7) for Virapap[®] and 28.8 (95% CI = 10.0-66.7) for SH. HPV-16 was the most common type detected in both cases and controls. Our results indicate that there is a very strong association between HPV 16, 18, 31, 33 and 35 and invasive cervical cancer and that this association is probably causal.

Human papillomaviruses (HPV) have emerged over the past 15 years as most likely to be the sexually transmitted aetiological factor in cervical cancer. Considerable advances have recently been made in the understanding of the molecular mechanisms by which certain HPV types might contribute to

UICC
Publication of the International Union Against Cancer
Publication de l'Union Internationale Contre le Cancer

TABLE II - ODDS RATIO (AND 95% CONFIDENCE INTERVALS) FOR THE ASSOCIATION BETWEEN INVASIVE CERVICAL CANCER AND HPV-DNA AS DETECTED BY VIRAPAP, SH AND PCR AMPLIFICATION

Hybridization assay	Spain				Colombia				Spain and Colombia	
	Cases (%)	Controls (%)	OR ¹	ORa ² (95% CI)	Cases (%)	Controls (%)	OR ¹	ORa ² (95% CI)	OR ¹	ORa ³ (95% CI)
Virapap (score)	(N = 250)	(N = 238)			(N = 186)	(N = 149)				
Negative (0-1)	58.9	97.5	1.0	1	69.6	99.3	1	1	1	1
Positive (2-5)	41.1	2.5	28.9	29.1 (10.3-81.9)	30.4	0.7	62.6	93.2 (12.6-687.3)	25.2	25.9 (10.0-66.7)
Southern hybridization	(N = 184)	(N = 158)			(N = 132)	(N = 143)				
Negative	63.6	94.3	1.0	1	75.0	95.8	1	1	1	1
Positive ⁴	36.4	5.7	9.9	8.8 (4.1-19.0)	25.0	4.2	7.5	9.6 (3.7-25.1)	6.9	6.8 (3.4-13.4)
Positive (SH) ⁵	31.2	0.0	—	—	22.0	1.4	19.0	23.4 (5.4-102.1)	55.9	53.9 (13.2-220.1)
PCR ⁶	(N = 142)	(N = 130)			(N = 87)	(N = 98)				
Negative	30.9	95.4	1.0	1	27.6	86.7	1.0	1	1	1
Positive	69.0	4.6	49.5	46.2 (18.5-115.1)	72.4	13.3	17.1	15.6 (6.9-34.7)	26.6	28.8 (15.7-52.6)

¹OR¹, adjusted for age and study area. ²ORa², adjusted for age, study area, number of sexual partners, education, age at first birth and history of a prior Pap smear. ³OR³, includes only those subjects tested by the 3 tests (cases = 227; controls = 224). ⁴SH-positive includes those samples which were positive under low-stringency and/or high-stringency conditions. ⁵SH-positive at high stringency only. ⁶PCR-positive includes those samples which were positive with type-specific probes and/or generic probes and excludes 45 samples negative for both β -globin and HPV-DNA.

TABLE III - HPV TYPES AS DETECTED BY PCR AMPLIFICATION IN CASES AND CONTROLS

HPV type	Colombia				Spain			
	Cases		Controls		Cases		Controls	
	n	%	n	%	n	%	n	%
16	44	(69.9)	9	(69.2)	65	(66.4)	4	(66.6)
18	5	(7.9)	2	(15.4)	5	(5.1)	—	—
31	5	(7.9)	—	—	5	(5.1)	1	(16.7)
33	—	—	—	—	5	(5.1)	1	(16.7)
35	3	(4.8)	1	(7.7)	2	(2.0)	—	—
Unknown	6	(9.5)	1	(7.7)	16	(16.3)	—	—
Total	63	(100.0)	13	(100.0)	98	(100.0)	6	(100.0)

TABLE IV - VIRAL LOAD OF HPV AND RISK OF CERVICAL CANCER

HPV features	Cases	Controls	OR ¹	95% CI
Virapap (score)				
Very weak positive (1)	27	10	1.0	
Positive (2-3)	59	3	8.6	(1.7-42.2)
Strong positive (4-5)	58	2	10.3	(1.7-64.2)
$p = 0.004^2$				
Southern blot (copy number/cell)				
< 0.2	3	0		
0.2-4.99	33	1		
≥ 5	45	1		

¹OR adjusted for age, study area. ²p-value for linear trend.

161 (67.6%) of the controls interviewed provided cytological specimens. In Colombia, the corresponding figures were: 186 (81.9%) cases and 149 (84.2%) controls agreed to participate, and 135 (72.6%) cases and 145 (97.3%) controls provided cytological specimens. The proportion of refusals in Spain was 11.5% among cases and 26.2% among controls and in Colombia it was 16.0% among cases and 19.8% among controls.

No significant differences were observed between those who provided cervical cells for HPV-DNA assays and those who did not, concerning the prevalence of the other risk factors under study. The possibility of selection bias is further discussed in Bosch *et al.* (1992).

As expected, the age distribution of the controls matched well the age distribution of the cervical cancer cases. In Colombia the mean age was 46.5 years in cases and 47.5 in

COHORT

- Measure Incidence
 - No of new cases only
- Measure relative risk , hazard ratio

$$\text{prevalence} = \frac{\text{ALL cases}}{\text{Population @ risk}}$$

$$\text{Incidence} = \frac{\text{New cases}}{\text{Population @ risk}}$$

Table 2: Hazard ratios for diabetes in children and youth

Characteristic	Birth to age 22 yr		Birth to age 12 yr		Age 12 to 22 yr	
	Unadjusted HR (95% CI)	Adjusted HR (95% CI)	Unadjusted HR (95% CI)	Adjusted HR (95% CI)	Unadjusted HR (95% CI)	Adjusted HR (95% CI)
Maternal gestational diabetes	1.88 (1.51–2.33)	1.77 (1.41–2.22)	1.52 (1.16–1.97)	1.43 (1.09–1.89)	2.89 (1.94–4.29)	2.53 (1.67–3.85)
Female sex*	1.18 (0.96–1.45)	1.23 (0.99–1.52)	0.95 (0.74–1.23)	0.99 (0.76–1.29)	1.74 (1.22–2.49)	1.86 (1.28–2.70)
Gestational age, wk†						
< 32	1.66 (0.89–3.13)	1.53 (0.78–3.0)	1.70 (1.15–2.64)	1.29 (0.47–3.50)	3.54 (1.56–8.05)	2.97 (1.19–7.42)
32 to 37	1.39 (0.96–2.0)	1.13 (0.74–1.72)	1.29 (0.48–3.46)	1.56 (0.97–2.50)	0.73 (0.32–1.65)	0.42 (0.15–1.18)
Birth weight category, g‡						
< 3000	1.31 (1.02–1.67)	1.24 (0.95–1.63)	1.30 (0.95–1.76)	1.17 (0.83–1.65)	1.30 (0.86–1.98)	1.34 (0.85–2.11)
> 4000	1.38 (1.01–1.88)	1.35 (0.98–1.85)	1.33 (0.90–1.97)	1.33 (0.87–1.98)	1.51 (0.91–2.53)	1.42 (0.84–2.41)
Ethnocultural background						
Non-European§	1.04¶ (0.79–1.35)	1.08** (0.82–1.41)	0.98 (0.71–1.37)	1.05 (0.76–1.47)	0.85 (0.54–1.33)	0.80 (0.51–1.27)
Deprivation level						
4 least deprived quintiles††	1.08 (0.83–1.4)	1.13 (0.86–1.47)	1.23 (0.88–1.72)	1.27 (0.90–1.79)	0.87 (0.57–1.31)	0.91 (0.59–1.34)
Previous pregnancy	1.12 (0.91–1.38)	1.04 (0.84–1.30)	1.0 (0.77–1.31)	0.96 (0.73–1.26)	2.02 (1.42–2.87)	1.74 (1.20–2.54)
Maternal autoimmune disease	1.22 (0.98–1.51)	1.14 (0.91–1.42)	1.27 (0.97–1.67)	1.21 (0.92–1.60)	1.34 (0.94–1.93)	1.20 (0.82–1.75)

Note: CI = confidence interval, HR = hazard ratio.
 *Reference group is boys.
 †Reference group for gestational age is 37 weeks or more.
 ‡Reference group for birth weight categories is 3000 to 4000 g.
 §European background is defined as birth in North America, Europe and Australia with European first language.
 ¶In univariate analyses, Indigenous origin versus European (unadjusted HR 3.42, 95% CI 1.70–6.9) and West Asian origin versus European (unadjusted HR 1.63, 95% CI 1.08–2.44) showed associations with offspring diabetes.
 **In multivariate analyses, Indigenous origin versus European (adjusted HR 3.49, 95% CI 1.67–7.27) and West Asian origin versus European (adjusted HR 1.68, 95% CI 1.11–2.53) also showed associations with offspring diabetes.
 ††Four least deprived quintiles compared with most deprived quintile classified by the Institut national de santé publique material deprivation index.¹⁸

RESEARCH

Gestational diabetes associated with incident diabetes in childhood and youth: a retrospective cohort study

Andrea L. Blotsky MD MSc, Elham Rahme PhD, Mourad Dahhou MSc, Meranda Nakhla MD MSc, Kaberi Dasgupta MD MSc

■ Cite as: CMAJ 2019 April 15;191:E410-7. doi: 10.1503/cmaj.181001

ABSTRACT

BACKGROUND: Indicators of childhood- and youth-onset diabetes may be useful for early detection of diabetes; there is a known association between composite exposure of parental type 2 diabetes and gestational diabetes mellitus with childhood- and youth-onset diabetes. We examined associations between gestational diabetes mellitus and incidence of childhood- and youth-onset diabetes in offspring.

METHODS: Using public health insurance administrative databases from Quebec, Canada, we randomly selected singleton live births with maternal gestational diabetes mellitus (1990–2007) and matched them 1:1 with singleton live births without gestational diabetes

mellitus. Follow-up was to Mar. 31, 2012. We examined associations of diabetes in offspring with maternal gestational diabetes mellitus through unadjusted and adjusted Cox proportional hazards models. In secondary analyses, we separately considered age groups ranging from birth to age 12 years, and age 12 to 22 years.

RESULTS: Incidence of pediatric diabetes (per 10 000 person-years) was higher in offspring born to mothers with gestational diabetes mellitus (4.52, 95% confidence interval [CI] 4.47–4.57) than in mothers without gestational diabetes mellitus (2.4, 95% CI 2.37–2.46). In an adjusted Cox proportional hazards model, maternal gestational diabetes

mellitus was associated with development of pediatric diabetes overall (birth to age 22 yr: hazard ratio [HR] 1.77, 95% CI 1.41–2.22), during childhood (birth to age 12 yr: HR 1.43, 95% CI 1.09–1.89), and in youth (age 12 to 22 yr: HR 2.53, 95% CI 1.67–3.85).

INTERPRETATION: Gestational diabetes mellitus is associated with incident diabetes in offspring during childhood and adolescence. Future studies are needed to examine longer-term outcomes in patients with pediatric diabetes with a maternal history of gestational diabetes mellitus, to ascertain how they compare with other patients with childhood- or youth-onset diabetes, in terms of disease severity and outcomes.

More than 90% of childhood- and youth-onset diabetes in Canada is type 1 diabetes.^{1,2} The peak prevalence of persistent autoantibodies to islet cell antigens occurs by age 2 years,³ with some individuals progressing to type 1 diabetes years later on in life. Indicators of higher likelihood of progression (e.g., recurrent enteroviral infections or absence of rotavirus vaccination,^{4,5} higher maternal or paternal age⁶) could potentially

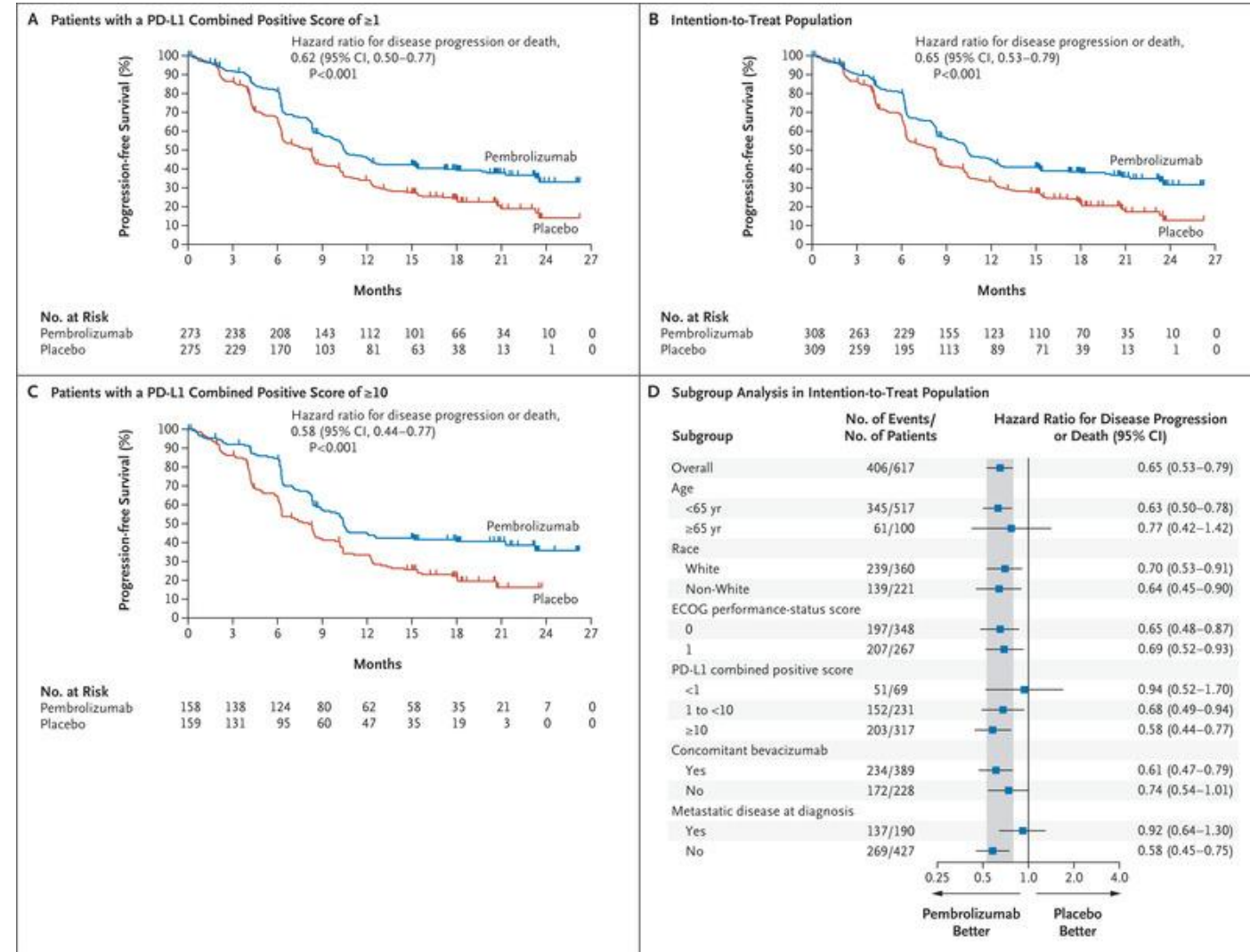
diabetes,⁹ showed a 6.2-fold higher incidence rate ratio (IRR) for childhood- and youth-onset diabetes with maternal type 1 diabetes and a 7.6-fold higher IRR with paternal type 1 diabetes.⁸ The IRR was 1.9 times higher for a composite exposure of maternal type 2 diabetes and gestational diabetes mellitus versus neither maternal condition, and 2.5 times as high with versus without paternal diabetes.

Pembrolizumab for Persistent, Recurrent, or Metastatic Cervical Cancer

Nicoletta Colombo, M.D., Ph.D., Coraline Dubot, M.D., Domenica Lorusso, M.D., Ph.D., M. Valeria Caceres, M.D., Ph.D., Kosei Hasegawa, M.D., Ph.D., Ronnie Shapira-Frommer, M.D., Krishnansu S. Tewari, M.D., Pamela Salaman, M.D., Edwin Hoyos Usta, M.D., Eduardo Yañez, M.D., Mahmut Gümüş, M.D., Mivael Olivera Hurtado de Mendoza, M.D., et al., for the KEYNOTE-826 Investigators*

CLINICAL TRIAL

Measure efficacy and effectiveness
One sided p value for non inferiority trial
Two sided p value for superiority trial



META ANALYSIS

- Highest hierarchy of data
- Forest plot
- Heterogeneity

Journal List > J Clin Oncol > PMC2645100

Journal of Clinical Oncology®
An American Society of Clinical Oncology Journal

J Clin Oncol. 2008 Dec 10; 26(35): 5802-5812.

Published online 2008 Nov 10. doi: [10.1200/JCO.2008.16.4368](https://doi.org/10.1200/JCO.2008.16.4368)

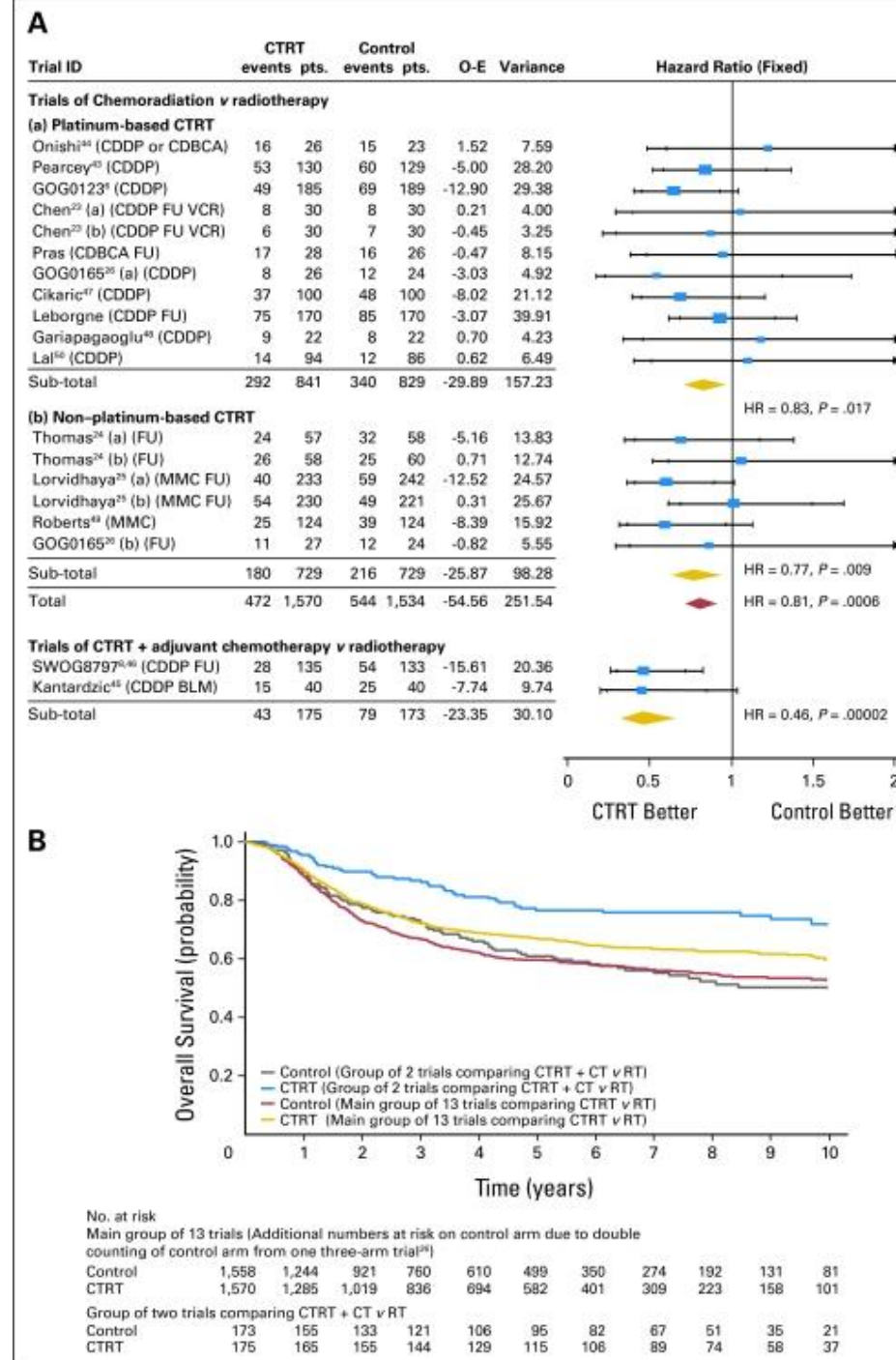
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PMID: [19001332](https://pubmed.ncbi.nlm.nih.gov/19001332/)

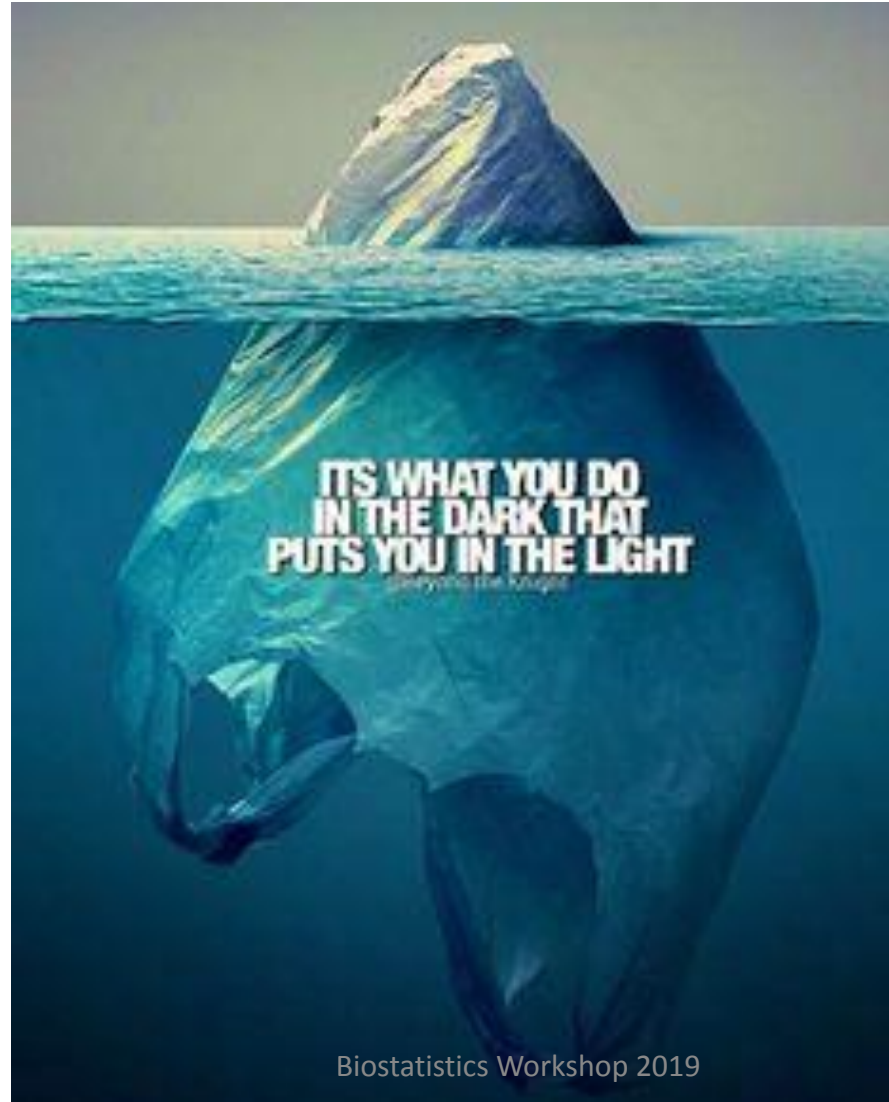
Reducing Uncertainties About the Effects of Chemoradiotherapy for Cervical Cancer: A Systematic Review and Meta-Analysis of Individual Patient Data From 18 Randomized Trials

Chemoradiotherapy for Cervical Cancer Meta-Analysis Collaboration

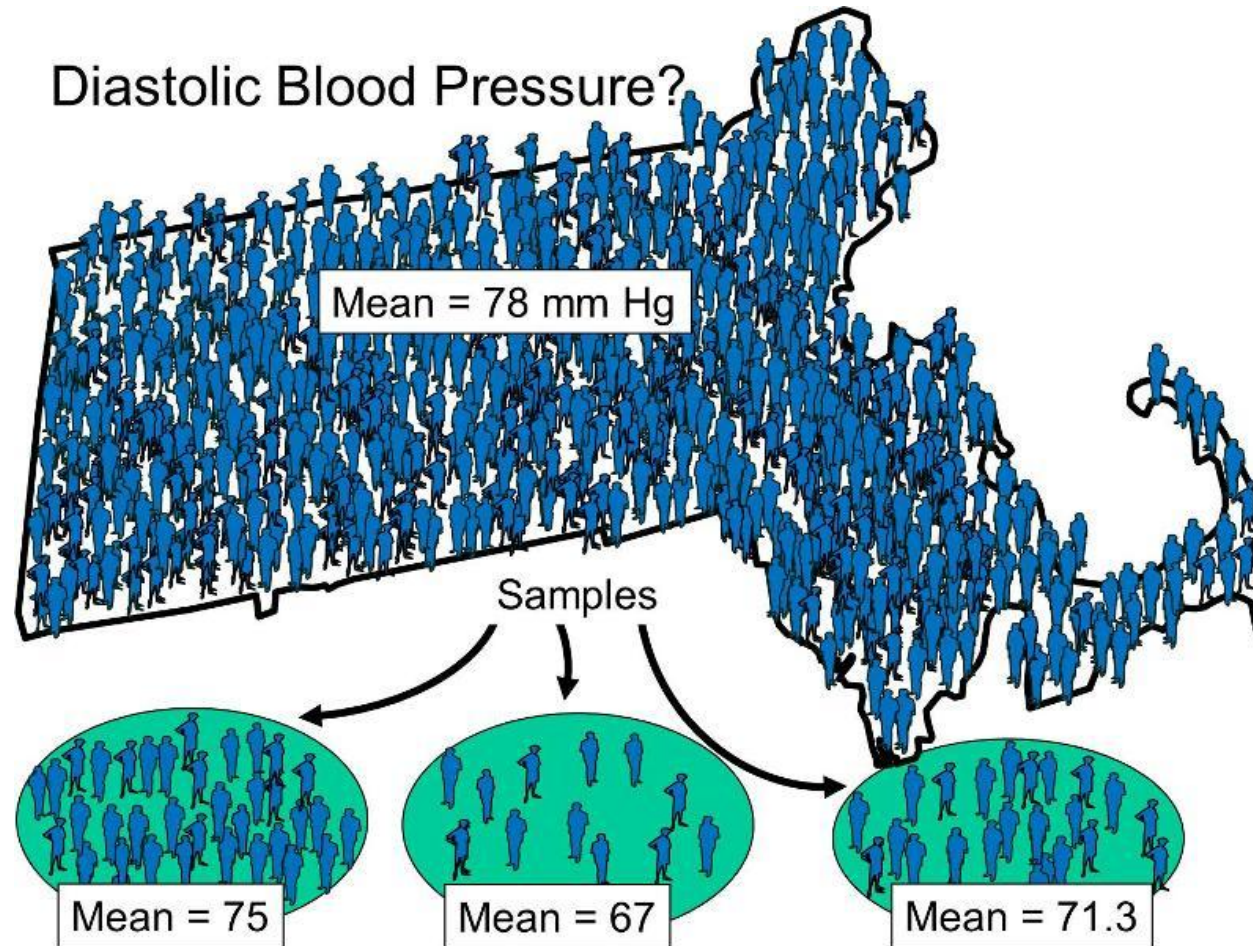
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Iceberg phenomenon

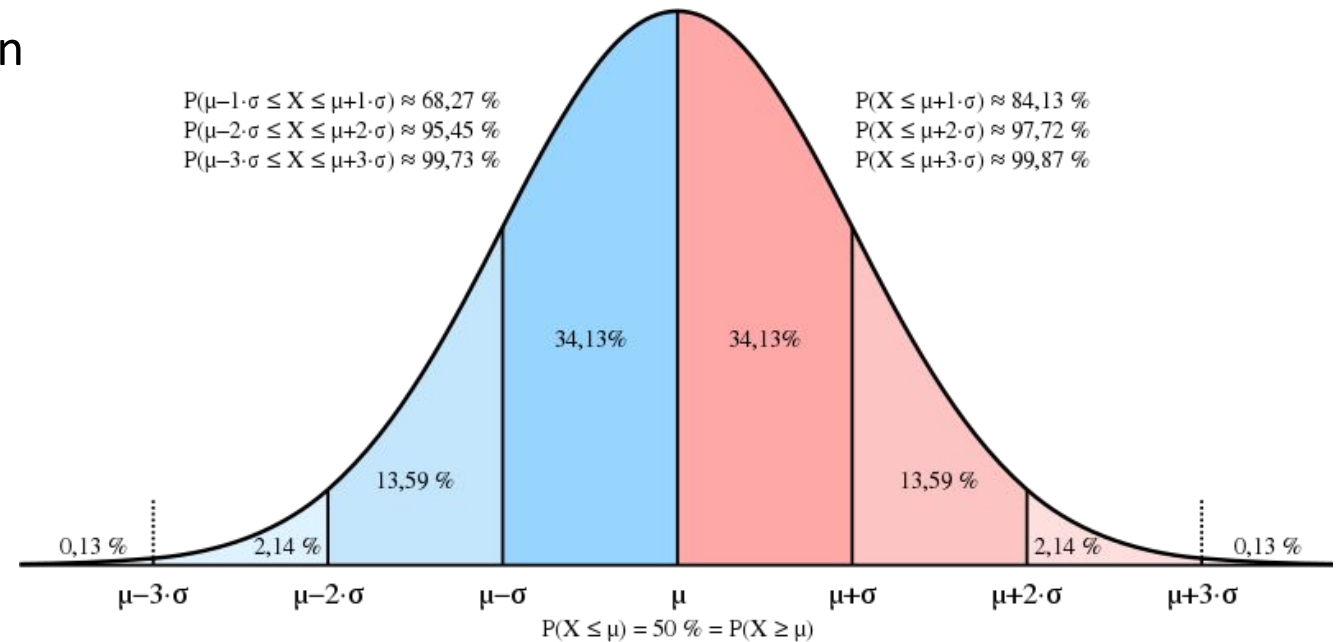


Statistic versus parameter



PRINCIPLE OF STATISTICS

- Sampling technique to represent population
 - Random sampling
 - Sample size
- Distribution of data
 - Distribution
 - Normal distribution
 - Central limit theorem
 - Other distribution
 - Assumption for most statistical inference
 - Assumption for many statistical test

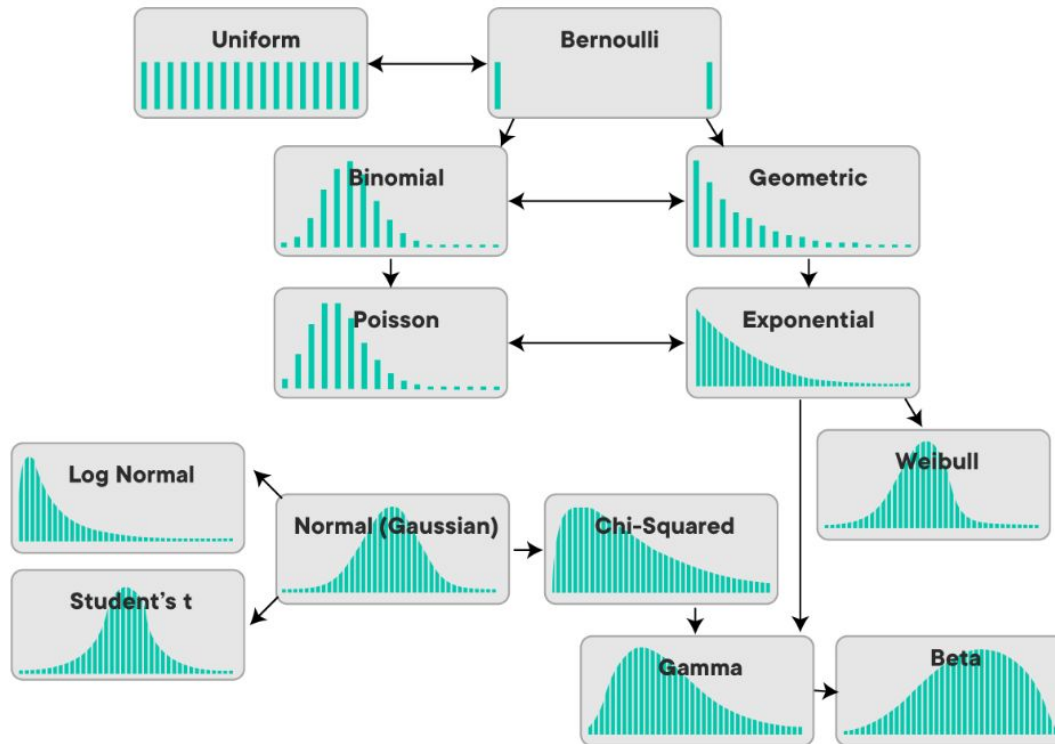


It is a capital mistake to theorize before one has data.

Sir Arthur Ignatius Conan Doyle

(1859 – 1930, Scottish physician and writer, most noted for his stories about Sherlock Holmes)

Distribution of data



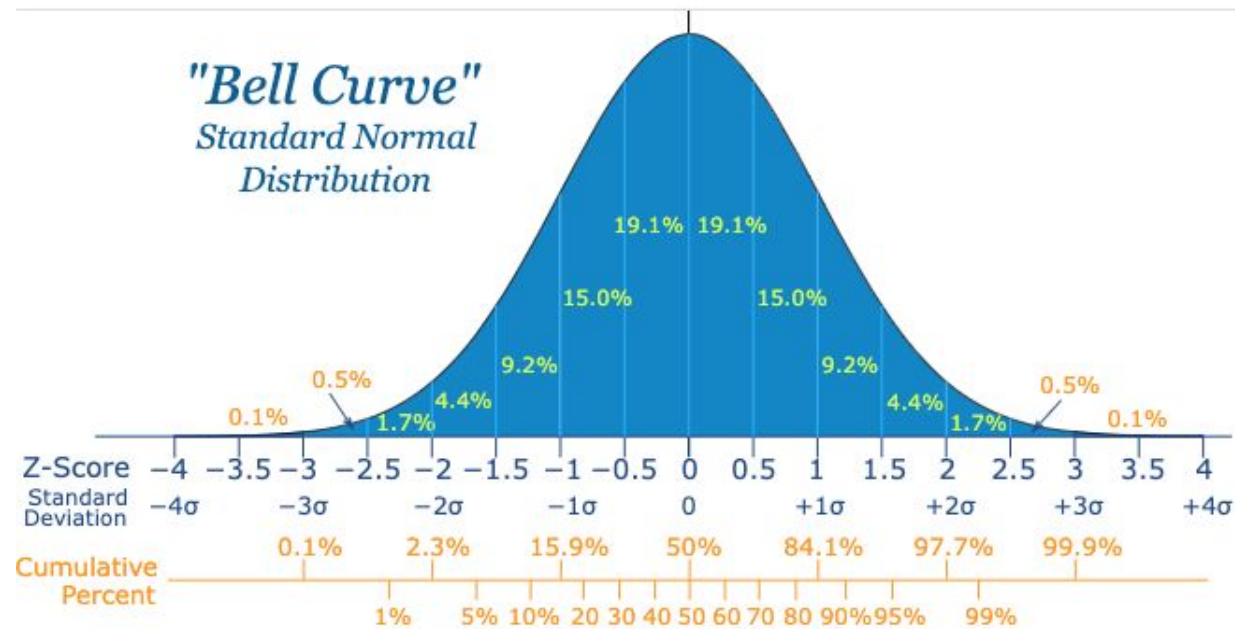
Applicable to numerical value

Discrete or Continuous data

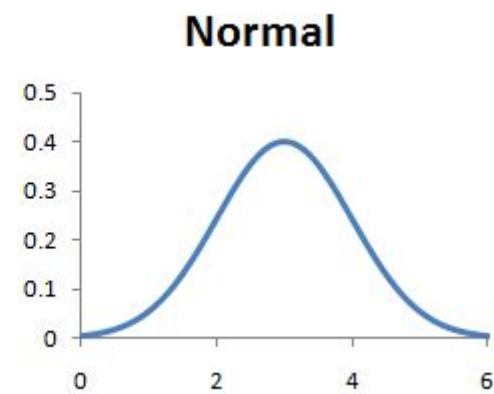
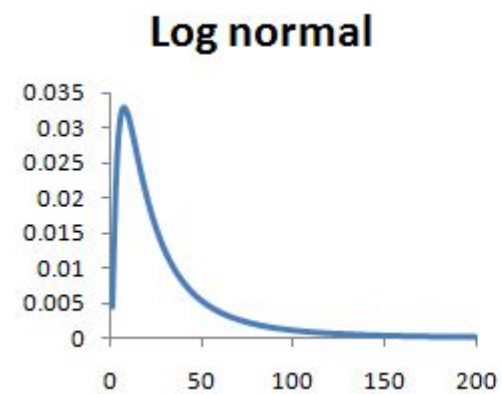
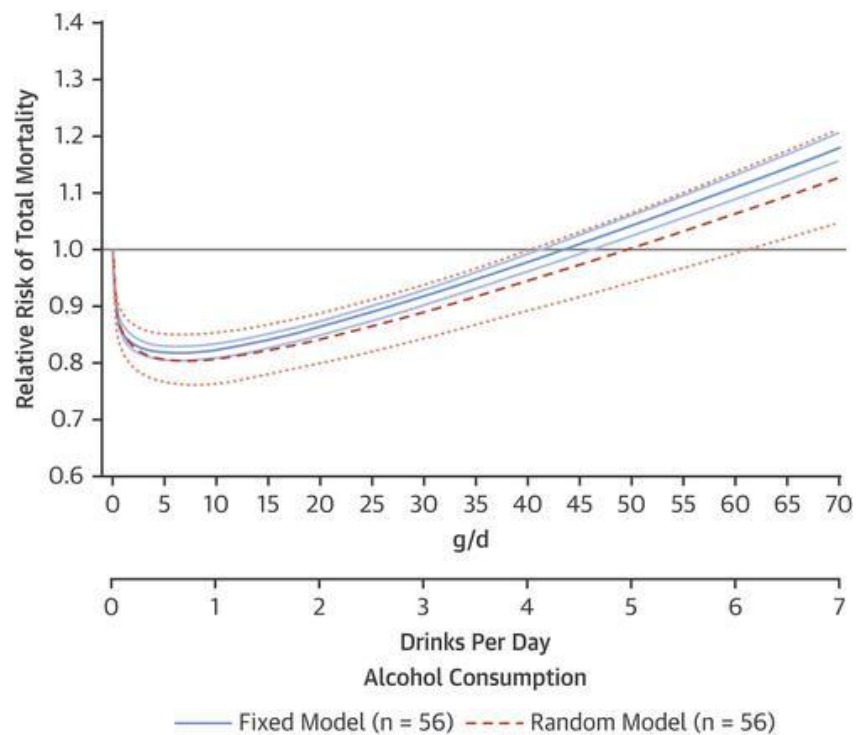
Discrete ~ *binomial, Poisson, Negative binomial, Multinomial etc*

Continuous ~ *Normal, t, F, Chi square*

Normal distribution



- Most natural phenomena observe this normal distribution
- Characteristics of normal distribution
 - It is "Bell-Shaped" and has a single peak at the center of the distribution
 - The arithmetic Mean, Median and Mode are equal.
 - The total area under the curve is 1.00
 - Half the area under the normal curve is to the right of this centre point and the other half to the left of it
 - It is Symmetrical about the mean
 - It is Asymptotic: The curve gets closer and closer to the X – axis but never actually touches it.
 - To put it another way, the tails of the curve extend indefinitely in both directions.
 - The location of a normal distribution is determined by the Mean, μ
 - The Dispersion or spread of the distribution is determined by the Standard Deviation, σ .



J curve - log-normal , $y = \ln(x)$



Descriptive statistics

- Numerical summaries of data sets
- Exploratory data analysis (EDA) to generate inferential
- In any study , before we answer our research questions, we explore the characteristics of the variables (e.g the age, gender, ethnicity).
- Use descriptive statistics
 - Categorical data
 - Continuous data

Descriptive statistics

- Categorical data
 - Frequencies
 - Percentages
 - Fractions and/or relative frequencies
 - Cross tabulations
- Continuous data
 - Measure of central tendency
 - Mean, median, mode
 - Arithmetic mean, geometric mean
 - Measure of dispersion
 - Standard deviations, variance, interquartile range, min, max

Table 1. Stage of Ovarian Cancer at Diagnosis and Survival at 5 Years.*

Tumor Stage	Patients with Tumor Stage at Diagnosis		5-Yr Survival
		percent	
I			
II			
III†	37		
A	8		83–90
B	44		65–71
C	3		
Outcome of initial surgery‡	5		47
No residual tumor	36		42
Residual disease			33
Optimal debulking (<2-cm nodules)	NA		63
Suboptimal debulking (≥2-cm nodules)	NA		33
IV	NA		25
	11		19

* Data are from Heintz et al.

† Stage III is divided into the pelvic and the peritoneal regions.

‡ Data are from Heintz et al.

Table 2. Adverse Events of Any Cause with an Incidence of 20% or More in Either Group (As-Treated Population).*

Event	Pembrolizumab Group (N=307)†		Placebo Group (N=309)†	
	Any Grade	Grade 3–5	Any Grade	Grade 3–5
Any event	305 (99.3)	251 (81.8)‡	307 (99.4)	232 (75.1)§
Anemia	188 (61.2)	93 (30.3)	165 (53.4)	83 (26.9)
Alopecia	173 (56.4)	0	179 (57.9)	0
Nausea	109 (35.5)	11 (3.6)	135 (43.7)	5 (1.6)
Diarrhea	88 (28.7)	1 (0.3)	92 (29.8)	8 (2.6)
Fatigue	87 (28.3)	2 (0.7)	84 (27.2)	14 (4.5)
Constipation	82 (26.7)	8 (2.6)	102 (33.0)	3 (1.0)
Arthralgia	81 (26.4)	8 (2.6)	80 (25.9)	4 (1.3)
Peripheral neuropathy	81 (26.4)	29 (9.4)	79 (25.6)	9 (2.9)
Vomiting	74 (24.1)	27 (8.8)	84 (27.2)	6 (1.9)
Hypertension	73 (23.8)	38 (12.4)	71 (23.0)	33 (10.7)
Urinary tract infection	72 (23.5)	3 (1.0)	80 (25.9)	25 (8.1)
Neutropenia	71 (23.1)	11 (3.6)	60 (19.4)	30 (9.7)
Peripheral sensory neuropathy	63 (20.5)	23 (7.5)	79 (25.6)	6 (1.9)
Asthenia	61 (19.9)		66 (21.4)	5 (1.6)
Thrombocytopenia			62 (20.1)	14 (4.5)

* Shown are adverse events that occurred while patients were receiving trial agents or within 30 days after the end of the trial treatment period (or, for serious events, within 90 days after the end of trial treatment or within 30 days if the patient initiated new anticancer therapy). The as-treated population included all the patients who underwent randomization and received at least one dose of pembrolizumab or placebo. Adverse events were classified according to the Medical Dictionary for Regulatory Activities, version 24.0.

† The assigned regimen in both groups also included paclitaxel, the investigator's choice of cisplatin or carboplatin, and bevacizumab.

‡ The maximum grade was grade 3 for 167 patients (54.4%), grade 4 for 70 patients (22.8%), and grade 5 for 14 patients (4.6%).

§ The maximum grade was grade 3 for 176 patients (57.0%), grade 4 for 42 patients (13.6%), and grade 5 for 14 patients (4.5%).

Table 1. Baseline Characteristics of the Patients.*

Characteristic	Open Surgery (N=312)	Minimally Invasive Surgery (N=319)
Age — yr	46.0±10.6	46.1±11.0
Body-mass index†	26.2±5.3	27.2±5.6
Histologic subtype — no. (%)		
Squamous-cell carcinoma	210 (67.3)	214 (67.1)
Adenocarcinoma	80 (25.6)	87 (27.3)
Adenosquamous carcinoma	6 (1.9)	9 (2.8)
Not reported	16 (5.1)	9 (2.8)
Stage of disease — no. (%)		
IA1: lymphovascular invasion	5 (1.6)	5 (1.6)
IA2	20 (6.4)	21 (6.6)
IB1	287 (92.0)	293 (91.8)
ECOG performance-status score — no. (%)‡		
0	289 (92.6)	292 (91.5)
1	23 (7.4)	27 (8.5)
Median length of hospital stay (range) — days	5 (0–69)§	3 (0–72)
Treatment received — no. (%)		
Open surgery	274 (87.8)	2 (0.6)
Minimally invasive surgery	8 (2.6)	289 (90.6)
Patient withdrew before surgery	19 (6.1)	12 (3.8)
Surgery was aborted	11 (3.5)	16 (5.0)

* Plus-minus values are means ±SD. Minimally invasive surgery indicates laparoscopic or robot-assisted radical hysterectomy, and open surgery indicates open abdominal radical hysterectomy. There were no significant differences in baseline characteristics between the assigned groups. Percentages may not total 100 because of rounding.

† The body-mass index is the weight in kilograms divided by the square of the height in meters.

‡ Performance-status scores on the Eastern Cooperative Oncology Group (ECOG) scale range from 0 to 4, with higher values indicating greater disability.

§ A zero length of stay in patients assigned to open surgery indicates patients who either withdrew before surgery or had surgery aborted and were discharged the same day.

Descriptive statistics

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INFERENCEAL STATISTICS

- Hypothesis testing (or statistical inference) is one of the major applications of biostatistics.
- To infer result to the population
- To test the research hypothesis in the form of statistical hypothesis

"Statistics are no substitute for judgment." Henry Clay

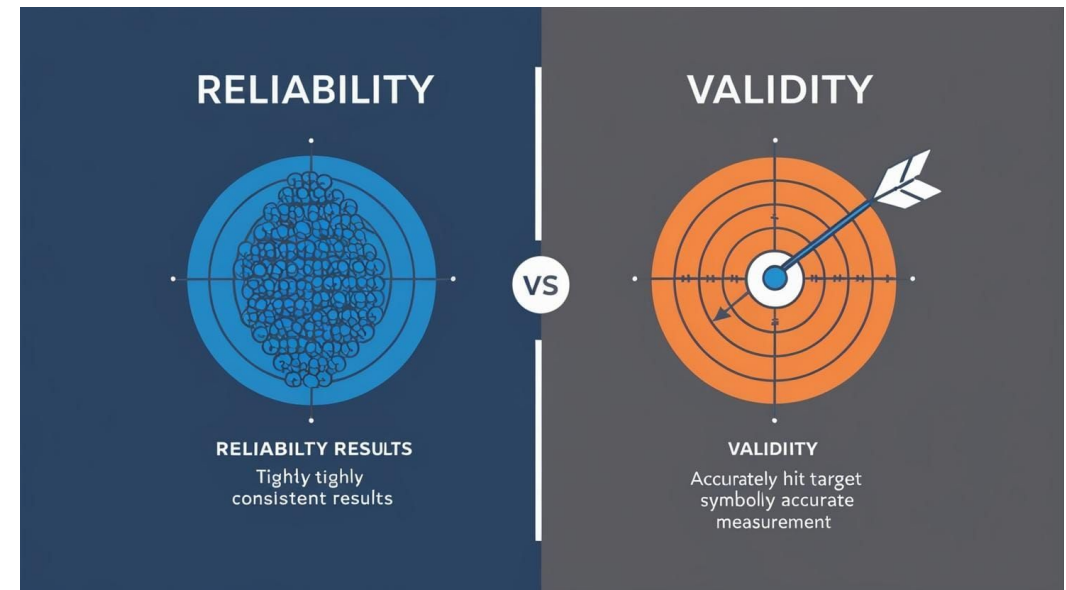
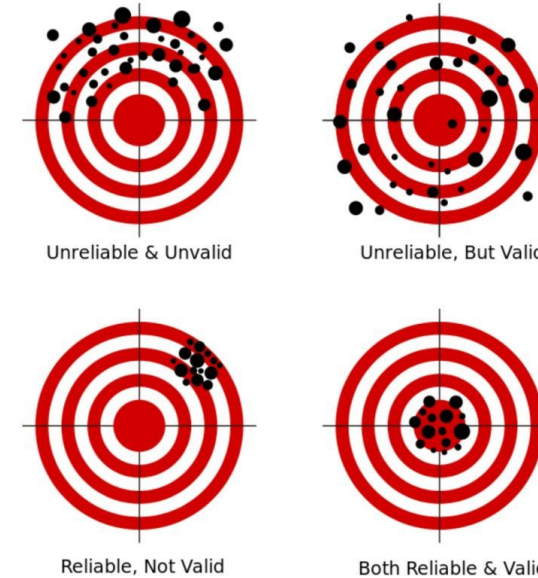
Ethical responsibilities

Statistical accuracy does not replace study validity

1. Scientific Integrity and Trial Design

A poorly designed trial is inherently unethical because it wastes resources and exposes human participants to potential risks without the possibility of gaining meaningful scientific knowledge.

- **Sample Size Sufficiency:** Biostatisticians must calculate a sample size that provides adequate statistical power to answer the research question. Enrolling too few patients means the trial cannot yield a conclusive result; enrolling too many unnecessarily exposes excess participants to experimental risks.
- **Minimizing Bias:** Biostatisticians are responsible for designing robust randomization schedules and blinding protocols. Proper execution prevents selection and observer bias, ensuring the trial's final data reflects true clinical outcomes.



2. Safeguarding Data Integrity

GCP demands that clinical trial data must be **accurate, complete, and verifiable**. Biostatisticians uphold this by

- **Pre-specifying the Statistical Analysis Plan (SAP):** The SAP must be finalized and signed off *before* data unblinding. This prevents "data dredging," "p-hacking," or changing endpoints retroactively to force a statistically significant result.
- **Handling Missing Data Objectively:** The strategy for dealing with dropouts and missing data points must be defined upfront to avoid biasing the results in favor of the investigational product.

3. Transparent Reporting and Avoiding Misrepresentation

The final responsibility lies in how data is communicated to regulatory bodies and the public

- **Reporting Negative Results:** Biostatisticians have an ethical obligation to report all findings honestly, regardless of whether they support the sponsor's commercial hypotheses. Suppressing negative data or selectively highlighting secondary endpoints violates the core foundation of ICH-GCP guidelines.

Extra reading on EQUATOR Network guideline and GCP / Good statistical practice

THANK YOU



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